GCSE CHEMISTRY

Higher Tier – Paper 1 style

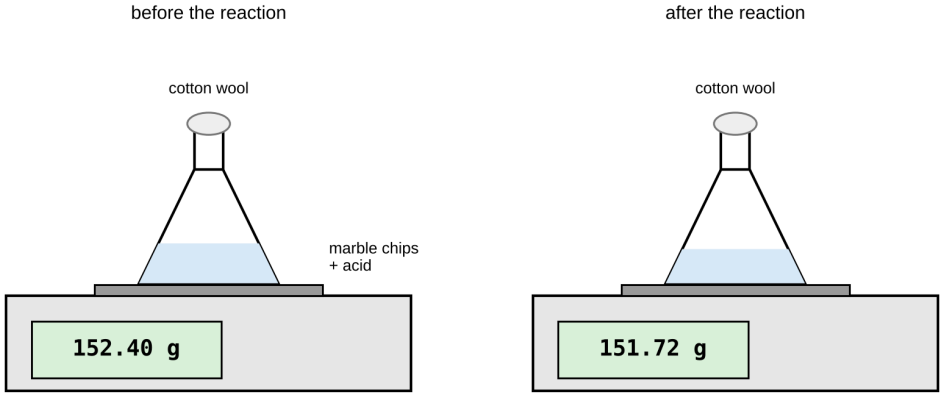
Topic 4.3: Quantitative chemistry

Practice questions in the style of AQA GCSE Chemistry (8462). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 140 minutes (roughly 1 minute per mark)
Materials	For this paper you must have: a ruler, a pencil, a calculator and the Periodic Table at the back of the question section.
Instructions	• Use black ink or a black ball-point pen (pencil for diagrams and graphs). • Answer **all** questions. • Answer the questions in the spaces provided. • Do all rough work on a separate sheet. • Cross through any work you do not want to be marked.
Information	• The maximum mark for this paper is **132**. • The marks for questions are shown in brackets. • You are allowed to use a calculator. • You are reminded of the need for good English and clear presentation in your answers. • Questions that ask you to *describe* or *explain* are marked for the science you include, so give each point its own sentence.

Name: _____ Date: _____ Mark: _____ / 132

0 1 A student reacts marble chips (calcium carbonate) with hydrochloric acid in an open conical flask. The flask is placed on a balance. A piece of cotton wool is placed in the neck of the flask. The readings on the balance are shown.



0 1 . 1 State the law of conservation of mass. **[1 mark]**

0 1 . 2 The mass shown on the balance decreases. Explain why. **[2 marks]**

0 1 . 3 Calculate the mass of carbon dioxide that escaped. **[1 mark]**

Answer = _____ g

0 1 . 4 What would the student observe about the mass if the experiment was carried out in a sealed container? Explain your answer. **[2 marks]**

0 2 A student heats 0.48 g of magnesium ribbon strongly in a crucible. She lets air in. The magnesium burns. The product is a white powder. The mass of the white powder is 0.80 g.

0 2 . 1 Explain why the mass of the white powder is greater than the mass of the magnesium. **[2 marks]**

0 2 . 2 Calculate the mass of oxygen that reacted. **[1 mark]**

Answer = _____ g

0 2 . 3 Balance the equations. **[4 marks]**

Equation
___ H_2 + ___ O_2 → ___ H_2O
___ Fe + ___ Cl_2 → ___ $FeCl_3$
___ Al + ___ O_2 → ___ Al_2O_3
___ C_3H_8 + ___ O_2 → ___ CO_2 + ___ H_2O

0 3 This question is about relative formula mass (M_r).
Relative atomic masses (A_r): H 1; C 12; N 14; O 16; Mg 24; Al 27; S 32; Cl 35.5; Ca 40; Fe 56.

0 3 . 1 Calculate the relative formula mass (M_r) of each substance. **[5 marks]**

Substance	**Working**	**M_r**
H_2O		
CO_2		
$MgCl_2$		
$Ca(OH)_2$		
$Al_2(SO_4)_3$		

0 3 . 2 Calculate the percentage by mass of magnesium in magnesium oxide, MgO. **[2 marks]**

Answer = _____ %

0 3 . 3 Calculate the percentage by mass of nitrogen in ammonium nitrate, NH_4NO_3. **[3 marks]**

Answer = _____ %

0 3 . 4 Iron is extracted from iron(III) oxide, Fe_2O_3. Calculate the maximum mass of iron that can be extracted from 1000 kg of iron(III) oxide. **[3 marks]**

Answer = _____ kg

0 4 This question is about the mole. Relative atomic masses (A_r): H 1; C 12; O 16; Na 23; S 32; Ca 40. The Avogadro constant is 6.02×10^{23} per mole.

0 4 . 1 Calculate the number of moles in 11.0 g of carbon dioxide, CO_2. **[2 marks]**

Answer = _____ mol

0 4 . 2 Calculate the mass of 0.20 mol of sodium hydroxide, NaOH. **[2 marks]**

Answer = _____ g

0 4 . 3 Calculate the number of atoms in 0.50 mol of carbon. Give your answer in standard form. **[2 marks]**

Answer = _____

0 4 . 4 A gas has a mass of 1.60 g. The sample contains 0.0250 mol of the gas. Calculate the relative formula mass of the gas and use A_r values to suggest what the gas is. **[3 marks]**

Answer = _____

0 5 This question is about calculating masses in reactions. Relative atomic masses (A_r): H 1; C 12; O 16; Mg 24; Al 27; Cl 35.5; Ca 40.

0 5 . 1 4.8 g of magnesium reacts with 3.2 g of oxygen to make 8.0 g of magnesium oxide. Use the masses to calculate the moles of Mg and O_2 and write a balanced equation for the reaction. **[4 marks]**

Answer = _____

0 5 . 2 Aluminium burns in oxygen: $4Al + 3O_2 \rightarrow 2Al_2O_3$ Calculate the mass of aluminium oxide formed when 5.4 g of aluminium burns completely. **[4 marks]**

Answer = _____ g

0 5 . 3 Calcium carbonate decomposes when heated: $CaCO_3 \rightarrow CaO + CO_2$ **[3 marks]**
Calculate the mass of calcium oxide made from 25 g of calcium carbonate.

Answer = _____ g

0 5 . 4 Magnesium reacts with hydrochloric acid: $Mg + 2HCl \rightarrow MgCl_2 + H_2$ **[3 marks]**
Calculate the mass of hydrogen produced from 12 g of magnesium.

Answer = _____ g

0 5 . 5 Propane burns: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ **[4 marks]**
Calculate the mass of carbon dioxide produced when 4.4 g of propane burns completely.

Answer = _____ g

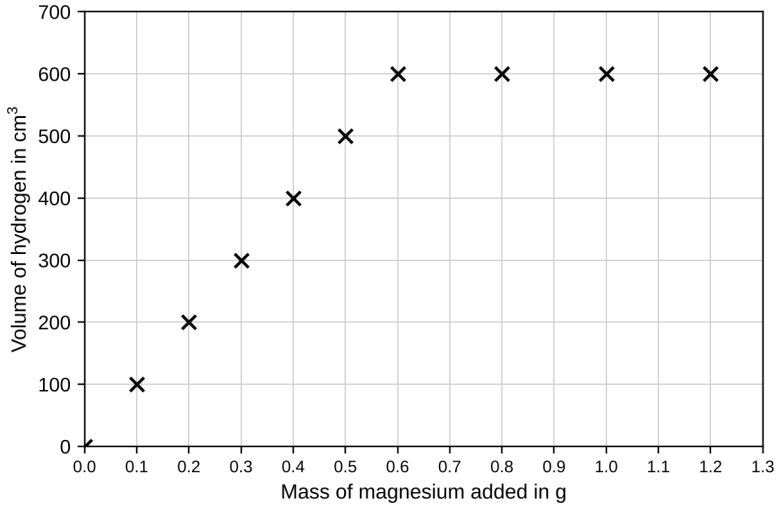
0 5 . 6 Both magnesium and calcium react with excess dilute acid in a 1 : 1 ratio with hydrogen ($M + 2HCl \rightarrow MCl_2 + H_2$). **[4 marks]**
Calculate which produces more hydrogen from 2.0 g of metal: magnesium (A_r 24) or calcium (A_r 40).

Answer = _____

0 6 A student adds different masses of magnesium to 50 cm^3 of hydrochloric acid of concentration 1.0 mol/dm^3. The student measures the volume of hydrogen gas made each time.

$Mg + 2HCl \rightarrow MgCl_2 + H_2$

The graph shows the results.



0 6 . 1 Which reactant was the limiting reactant when 0.3 g of magnesium was added? **[1 mark]**

0 6 . 2 The volume of hydrogen stops increasing when more than 0.60 g of magnesium is added. **[2 marks]**

Explain why.

0 6 . 3 Calculate the number of moles of hydrochloric acid in 50 cm^3 of 1.0 mol/dm^3 solution. **[2 marks]**

Answer = _____ mol

0 6 . 4 Use the equation to calculate the mass of magnesium that reacts with 0.050 mol of hydrochloric acid. Show that your answer agrees with the graph. **[3 marks]**

Answer = _____ g

0 6 . 5 Industrial processes often use an excess of one reactant. Give one reason why. **[1 mark]**

0 6 . 6 A student adds 1.2 g of magnesium to 100 cm^3 of 0.50 mol/dm^3 hydrochloric acid. **[4 marks]**
Calculate the moles of each reactant, state the limiting reactant and calculate the mass of hydrogen produced.

Answer = _____

0 7 This question is about the concentration of solutions. Relative atomic masses (A_r): H 1; O 16; Na 23; S 32.

0 7 . 1 20 g of a salt is dissolved in water to make 250 cm^3 of solution. **[2 marks]**
Calculate the concentration of the solution in g/dm^3.

Answer = _____ g/dm^3

0 7 . 2 A solution of sodium hydroxide has a concentration of 40 g/dm^3. Calculate the mass of sodium hydroxide in 500 cm^3 of the solution. **[2 marks]**

Answer = _____ g

0 7 . 3 4.0 g of sodium hydroxide is dissolved in water to make 250 cm^3 of solution. Calculate the concentration in mol/dm^3. **[3 marks]**

Answer = _____ mol/dm³

0 7 . 4 A solution of sodium hydroxide has a concentration of 80 g/dm^3. Calculate its concentration in mol/dm^3. **[2 marks]**

Answer = _____ mol/dm³

0 7 . 5 Calculate the mass of sulfuric acid, H_2SO_4, in 25.0 cm^3 of a 0.50 mol/dm^3 solution. **[3 marks]**

Answer = _____ g

0 8 This question is about percentage yield and atom economy. Relative atomic masses (A_r): C 12; O 16; Ca 40; Fe 56.

0 8 . 1 A student heats 25.0 g of calcium carbonate: $CaCO_3 \rightarrow CaO + CO_2$ **[2 marks]**
The theoretical yield of calcium oxide is 14.0 g. The student collects 11.2 g of calcium oxide.
Calculate the percentage yield.

Answer = _____ %

0 8 . 2 Give **two** reasons why the percentage yield of a reaction is less than 100%. **[2 marks]**

0 8 . 3 Iron is made by heating iron oxide with carbon: $Fe_2O_3 + 3C \rightarrow 2Fe + 3CO$ **[3 marks]**
Calculate the atom economy for making iron in this reaction.

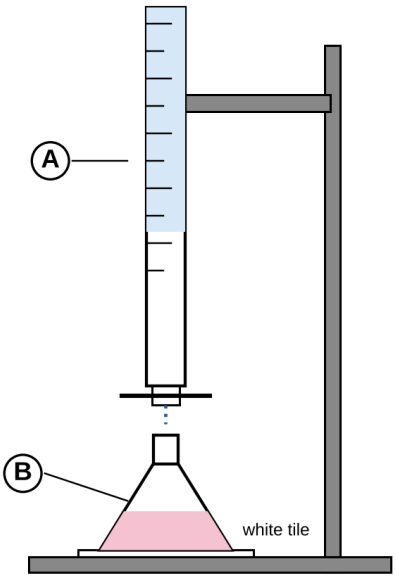
Answer = _____ %

0 8 . 4 Explain why it is important for companies to use reactions with a high atom economy. **[2 marks]**

0 8 . 5 Hydrogen peroxide decomposes: $2H_2O_2 \rightarrow 2H_2O + O_2$ **[3 marks]**
Calculate the atom economy for making oxygen. (A_r: H 1, O 16)

Answer = _____ %

0 9 A student carries out a titration to find the concentration of sodium hydroxide solution. She uses a pipette to measure 25.0 cm^3 of sodium hydroxide into a conical flask, adds a few drops of indicator and adds 0.100 mol/dm^3 hydrochloric acid from a burette. The apparatus is shown.



0 9 . 1 Name the pieces of apparatus labelled **A** and **B**. **[2 marks]**

0 9 . 2 Name a suitable indicator and give the colour change at the end point. **[2 marks]**

0 9 . 3 The table shows the student's results. **[2 marks]**
Calculate the mean titre using only concordant results.

Titration	**Rough**	**1**	**2**	**3**	**4**
Titre in cm^3	25.30	23.90	24.60	23.85	23.95

Answer = _____ cm^3

0 9 . 4 $NaOH + HCl \rightarrow NaCl + H_2O$ **[4 marks]**

Calculate the concentration of the sodium hydroxide solution in mol/dm^3.
Use 25.0 cm^3 of NaOH and 23.90 cm^3 of 0.100 mol/dm^3 HCl.

Answer = _____ mol/dm³

0 9 . 5 In a second titration 25.0 cm^3 of 0.200 mol/dm^3 sodium hydroxide is neutralised by 20.0 cm^3 of sulfuric acid. **[6 marks]**

$2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + 2H_2O$

Calculate the concentration of the sulfuric acid in mol/dm^3 and then in g/dm^3. (M_r of H_2SO_4 = 98)

Answer = _____

1 0 This question is about the volumes of gases. One mole of any gas occupies 24 dm^3 at room temperature and pressure. Relative atomic masses (A_r): H 1; C 12; O 16; Mg 24; Ca 40.

1 0 . 1 Calculate the volume of 0.25 mol of carbon dioxide at room temperature and pressure. **[1 mark]**

Answer = _____ dm³

1 0 . 2 10 g of calcium carbonate reacts with excess hydrochloric acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$ **[3 marks]**
Calculate the volume of carbon dioxide produced at room temperature and pressure.

Answer = _____ dm^3

1 0 . 3 0.12 g of magnesium reacts with excess acid. Calculate the volume of hydrogen made. Give your answer in cm^3. **[3 marks]**

Answer = _____ cm^3

1 0 . 4 Propane burns: $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$ **[2 marks]**
What volume of oxygen is needed to burn 4.0 dm^3 of propane completely?
Both gases are at the same temperature and pressure.

Answer = _____ dm^3

1 0 . 5 In the Haber process: $N_2 + 3H_2 \rightarrow 2NH_3$ **[2 marks]**
What volume of ammonia forms from 600 dm^3 of hydrogen if all the hydrogen reacts?

Answer = _____ dm^3

1 1 This question is about measurements and experiments.

1 1 . 1 The student's concordant titres were 23.90, 23.85 and 23.95 cm^3. **[2 marks]**
Calculate the mean titre and the uncertainty of the mean (uncertainty = range ÷ 2).

Answer = _____

1 1 . 2 The student uses a pipette to measure 25.0 cm^3 of sodium hydroxide, not a measuring cylinder. **[1 mark]**
Suggest why.

_____

_____

1 1 . 3 A sample contains 0.0100 mol of a substance. Calculate the number of particles in the sample. Give your answer in standard form. **[2 marks]**

Answer = _____

1 1 . 4 *A student heats magnesium ribbon in a crucible with a lid, and finds that the mass increases. **[6 marks]**
Describe how the student could do this experiment and explain why the mass increases.

_____

_____

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END OF QUESTIONS

The Periodic Table of the Elements (elements 1–56)

1 **H** 1																4 **He** 2
7 **Li** 3	9 **Be** 4											11 **B** 5	12 **C** 6	14 **N** 7	16 **O** 8	19 **F** 9	20 **Ne** 10
23 **Na** 11	24 **Mg** 12											27 **Al** 13	28 **Si** 14	31 **P** 15	32 **S** 16	35.5 **Cl** 17	40 **Ar** 18
39 **K** 19	40 **Ca** 20	45 **Sc** 21	48 **Ti** 22	51 **V** 23	52 **Cr** 24	55 **Mn** 25	56 **Fe** 26	59 **Co** 27	59 **Ni** 28	63.5 **Cu** 29	65 **Zn** 30	70 **Ga** 31	73 **Ge** 32	75 **As** 33	79 **Se** 34	80 **Br** 35	84 **Kr** 36
85.5 **Rb** 37	88 **Sr** 38	89 **Y** 39	91 **Zr** 40	93 **Nb** 41	96 **Mo** 42	[98] **Tc** 43	101 **Ru** 44	103 **Rh** 45	106 **Pd** 46	108 **Ag** 47	112 **Cd** 48	115 **In** 49	119 **Sn** 50	122 **Sb** 51	128 **Te** 52	127 **I** 53	131 **Xe** 54
133 **Cs** 55	137 **Ba** 56	57–71															

Group 1 → 18: left to right. Periods: top to bottom.

Relative atomic mass is shown above each symbol and atomic (proton) number below. Elements 57–71 are not shown.

MARK SCHEME – Topic 4.3: Quantitative chemistry

How to use this mark scheme: mark your own work in a different colour. A semicolon (;) separates marking points and each is worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed to gain the mark. **Allow** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. Marks are for the *science*, not the wording.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	no atoms are made or destroyed in a chemical reaction / the total mass of products equals the total mass of reactants	Do not accept 'mass cannot change' without 'in a closed system / total mass of products = reactants'.	1	AO1 4.3.1.1
01.2	carbon dioxide gas is made / given off ; and escapes from the flask (through the cotton wool) so its mass is no longer measured ;	Do not accept 'the marble chips dissolve so the mass drops'. The atoms are still there in the solution.	2	AO2 4.3.1.1
01.3	152.40 − 151.72 = 0.68 g		1	AO2 4.3.1.1
01.4	the mass stays the same / no change ; no substances can escape / the gas stays in the flask so the total mass is unchanged ;	Safety note: do not seal a flask where gas is produced (pressure build-up). Real experiments use a gas syringe.	2	AO2 4.3.1.1
02.1	magnesium reacts with oxygen (from the air) ; to make magnesium oxide, so the oxygen's mass is added ;		2	AO2 4.3.1.1
02.2	0.80 − 0.48 = 0.32 g		1	AO2 4.3.1.1
02.3	$2H_2 + O_2 \rightarrow 2H_2O$; $2Fe + 3Cl_2 \rightarrow 2FeCl_3$; $4Al + 3O_2 \rightarrow 2Al_2O_3$; $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$;	One mark for each equation. Balance by changing big numbers in front only, never the small numbers in formulae.	4	AO2 4.3.1.2
03.1	$H_2O = 18$; $CO_2 = 44$; $MgCl_2 = 24 + (2 \times 35.5) = 95$; $Ca(OH)_2 = 40 + 2 \times (16 + 1) = 74$; $Al_2(SO_4)_3 = 54 + 96 + 192 = 342$;	Common error: forgetting to multiply everything inside the bracket by the number outside. One mark each.	5	AO2 4.3.1.3
03.2	M_r of MgO = 40 ; (24 ÷ 40) × 100 = 60 (%) ;		2	AO2 4.3.1.3
03.3	$M_r = 14 + 4 + 14 + 48 = 80$; mass of nitrogen = 2 × 14 = 28 ; (28 ÷ 80) × 100 = 35 (%) ;	Common error: using only one N atom. There are **two** nitrogen atoms.	3	AO2 4.3.1.3
03.4	M_r of Fe_2O_3 = 160 ; mass of iron in one formula unit = 112, so 112/160 ; (112 ÷ 160) × 1000 = 700 kg ;		3	AO2 4.3.1.3
04.1	$M_r = 44$; 11.0 ÷ 44 = 0.25 (mol) ;		2	AO2 4.3.2.1
04.2	$M_r = 40$; 0.20 × 40 = 8.0 g ;	Formula: mass = moles × M_r.	2	AO2 4.3.2.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.3	$0.50 \times 6.02 \times 10^{23}$; 3.01×10^{23} ;		2	AO2 4.3.2.1
04.4	M_r = mass ÷ moles = 1.60 ÷ 0.0250 ; = 64 ; sulfur dioxide / SO_2 (32 + 2 × 16 = 64) ;		3	AO3 4.3.2.1
05.1	moles Mg = 4.8 ÷ 24 = 0.20 ; moles O_2 = 3.2 ÷ 32 = 0.10 ; ratio 0.20 : 0.10 = 2 : 1 ; $2Mg + O_2 \rightarrow 2MgO$;	Higher tier skill: moles give the ratio in the equation.	4	AO2 4.3.2.2
05.2	moles Al = 5.4 ÷ 27 = 0.20 ; moles Al_2O_3 = 0.20 × 2/4 = 0.10 ; M_r Al_2O_3 = 102 ; mass = 0.10 × 102 = 10.2 g ;		4	AO2 4.3.2.2
05.3	moles $CaCO_3$ = 25 ÷ 100 = 0.25 ; moles CaO = 0.25 (1 : 1) ; mass CaO = 0.25 × 56 = 14 g ;		3	AO2 4.3.2.2
05.4	moles Mg = 12 ÷ 24 = 0.50 ; moles H_2 = 0.50 (1 : 1) ; mass H_2 = 0.50 × 2 = 1.0 g ;	Common error: using 1 for the M_r of hydrogen gas. H_2 has M_r = 2.	3	AO2 4.3.2.2
05.5	M_r C_3H_8 = 44 ; moles propane = 4.4 ÷ 44 = 0.10 ; moles CO_2 = 0.10 × 3 = 0.30 ; mass CO_2 = 0.30 × 44 = 13.2 g ;		4	AO2 4.3.2.2
05.6	moles Mg = 2.0 ÷ 24 = 0.083 ; moles Ca = 2.0 ÷ 40 = 0.050 ; H_2 moles equal metal moles (1 : 1) ; magnesium produces more hydrogen ;	Same mass does **not** mean the same moles. Smaller A_r gives more moles.	4	AO3 4.3.2.2
06.1	magnesium		1	AO2 4.3.2.3
06.2	all of the hydrochloric acid has reacted / the acid is the limiting reactant ; so no more hydrogen can be made, however much more magnesium is added ;		2	AO3 4.3.2.3
06.3	50 cm^3 = 0.050 dm^3 ; moles = 1.0 × 0.050 = 0.050 mol ;		2	AO2 4.3.4
06.4	Mg : HCl = 1 : 2 so moles Mg = 0.025 ; mass = 0.025 × 24 = 0.60 g ; the graph levels off at 0.60 g ;		3	AO3 4.3.2.3
06.5	to make sure all of the other (more expensive) reactant is used up / to increase the yield		1	AO3 4.3.2.3
06.6	moles Mg = 1.2 ÷ 24 = 0.050 ; moles HCl = 0.50 × 0.100 = 0.050 ; 0.050 mol HCl needs only 0.025 mol Mg, so HCl is limiting ; moles H_2 = 0.025, mass = 0.025 × 2 = 0.050 g ;	Work out how much of one reactant is needed by the other using the 1 : 2 ratio. The reactant that runs out is limiting.	4	AO3 4.3.2.3
07.1	250 cm^3 = 0.25 dm^3 ; 20 ÷ 0.25 = 80 g/dm^3 ;	Common error: forgetting to convert cm^3 to dm^3 (divide by 1000).	2	AO2 4.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	500 cm^3 = 0.5 dm^3 ; 40 × 0.5 = 20 g ;		2	AO2 4.3.3
07.3	moles NaOH = 4.0 ÷ 40 = 0.10 ; 250 cm^3 = 0.25 dm^3 ; 0.10 ÷ 0.25 = 0.40 mol/dm^3 ;		3	AO2 4.3.4
07.4	80 ÷ 40 ; = 2.0 mol/dm^3 ;		2	AO2 4.3.4
07.5	moles = 0.50 × 0.0250 = 0.0125 mol ; M_r H_2SO_4 = 98 ; mass = 0.0125 × 98 = 1.2(25) g ;	Allow 1.2 or 1.23 g.	3	AO2 4.3.4
08.1	(11.2 ÷ 14.0) × 100 ; = 80 % ;	Percentage yield = actual yield ÷ theoretical yield × 100.	2	AO2 4.3.5
08.2	the reaction is reversible / does not go to completion ; some product is lost when it is separated / transferred ; some reactants react in other (side) reactions (any two) ;	Any two.	2	AO1 4.3.5
08.3	M_r of reactants = 160 + (3 × 12) = 196 ; M_r of desired product = 2 × 56 = 112 ; (112 ÷ 196) × 100 = 57.1 % ;	Atom economy = M_r of desired product ÷ sum of M_r of all reactants × 100.	3	AO2 4.3.6
08.4	less waste / fewer unwanted products ; conserves raw materials / more sustainable / lower costs ;		2	AO3 4.3.6
08.5	M_r of $2H_2O_2$ = 2 × 34 = 68 ; M_r of O_2 = 32 ; (32 ÷ 68) × 100 = 47.1 % ;		3	AO2 4.3.6
09.1	A: burette ; B: conical flask ;		2	AO1 4.3.4
09.2	methyl orange / phenolphthalein ; yellow to red (methyl orange) / pink to colourless (phenolphthalein) ;	The colour change must match the indicator named and the acid being added to alkali.	2	AO1 4.3.4
09.3	concordant results are 23.90, 23.85 and 23.95 (within 0.10 cm^3) ; mean = 23.90 cm^3 ;	Ignore the rough titre (25.30) and the 24.60 result.	2	AO2 4.3.4
09.4	moles HCl = 0.100 × 0.02390 = 0.00239 ; 1 : 1 ratio so moles NaOH = 0.00239 ; concentration = 0.00239 ÷ 0.0250 ; = 0.0956 mol/dm^3 ;	Allow 0.096 mol/dm^3.	4	AO2 4.3.4
09.5	moles NaOH = 0.200 × 0.0250 = 0.00500 ; ratio NaOH : H_2SO_4 = 2 : 1 ; moles H_2SO_4 = 0.00250 ; concentration = 0.00250 ÷ 0.0200 = 0.125 mol/dm^3 ; g/dm^3 = 0.125 × 98 ; = 12.25 g/dm^3 ;	Grade 8–9. Do not forget the 2 : 1 ratio. Allow 12.3 g/dm^3.	6	AO2 4.3.4
10.1	0.25 × 24 = 6.0 dm^3		1	AO2 4.3.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
10.2	moles $CaCO_3$ = 10 ÷ 100 = 0.10 ; moles CO_2 = 0.10 ; volume = 0.10 × 24 = 2.4 dm^3 ;		3	AO2 4.3.7
10.3	moles Mg = 0.12 ÷ 24 = 0.0050 ; volume = 0.0050 × 24 = 0.12 dm^3 ; = 120 cm^3 ;	1 dm^3 = 1000 cm^3.	3	AO2 4.3.7
10.4	volume ratio is the same as the mole ratio (Avogadro) so C_3H_8 : O_2 = 1 : 5 ; 4.0 × 5 = 20 dm^3 ;		2	AO2 4.3.7
10.5	H_2 : NH_3 = 3 : 2 ; 600 × 2/3 = 400 dm^3 ;		2	AO2 4.3.7
11.1	mean = 23.90 cm^3 ; range = 0.10 so uncertainty = ±0.05 cm^3 ;		2	AO2 4.3.8
11.2	a pipette is more accurate / has a smaller uncertainty (than a measuring cylinder)		1	AO3 4.3.8
11.3	0.0100 × 6.02 × 10^{23} ; 6.02 × 10^{21} ;		2	AO2 4.3.2.1
11.4	**Level 3 (5–6 marks)**: A clear, detailed and logically ordered answer that covers all of the key areas, with correct scientific reasoning linking the ideas together. **Level 2 (3–4 marks)**: Some of the key areas are covered with some correct reasoning, but the answer lacks detail, balance or logical order in places. **Level 1 (1–2 marks)**: Simple, separate statements with little or no linking or reasoning. **0 marks**: No relevant content. **Indicative content** (credit any other correct science): • weigh the empty crucible with its lid, then weigh it again with a piece of magnesium ribbon • heat strongly with a Bunsen burner, lifting the lid slightly now and then to let in air (oxygen) without letting smoke escape • continue until all the magnesium has turned to white ash (magnesium oxide) • allow to cool, then weigh the crucible, lid and contents again • the mass increases because magnesium combines with oxygen from the air to form magnesium oxide • mass of oxygen added = final mass − starting mass; no atoms are lost, so the law of conservation of mass is obeyed		6	AO1 4.3.1.1

Where marks are lost in Topic 4.3

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Units first:** convert cm^3 to dm^3 by dividing by 1000 before using concentration. This is the most common slip in titration and concentration questions.
- ☐ **Moles from mass:** moles = mass ÷ M_r. Use the M_r of the *whole* formula (H_2 = 2, not 1; O_2 = 32, not 16).
- ☐ **Ratio step:** always write the mole ratio from the balanced equation before using it (e.g. 2 : 1). Never skip it in a 4-mark calculation.
- ☐ **Equal mass is not equal moles.** A smaller A_r gives more moles for the same mass.
- ☐ **Limiting reactant:** work out the moles of both, divide by the equation ratio, and the smaller one is limiting. The limiting reactant decides how much product forms.
- ☐ **Percentage yield and atom economy are different:** yield uses actual ÷ theoretical masses; atom economy uses M_r of desired product ÷ M_r of all reactants.
- ☐ **Titrations:** use only concordant titres (within 0.10 cm^3) for the mean. Ignore the rough titre.
- ☐ **Gas volumes:** 24 dm^3 per mole at room temperature and pressure. Convert to cm^3 if the question asks (× 1000).
- ☐ **Balancing equations:** change only the numbers in front of formulae. Tick off each element on both sides.
- ☐ **Conservation of mass:** if the mass seems to change, say what gas escaped or what gas (oxygen) was gained. Use the words 'no atoms are made or destroyed'.
- ☐ **Show every step.** You can gain marks for method even with a wrong final answer. Give the answer to the number of significant figures in the data.